

Chapter 34

Creating and Working with Timelines

In this chapter, you'll learn how to create and modify the timelines into which you edit clips to create the edited sequences that are your programs.

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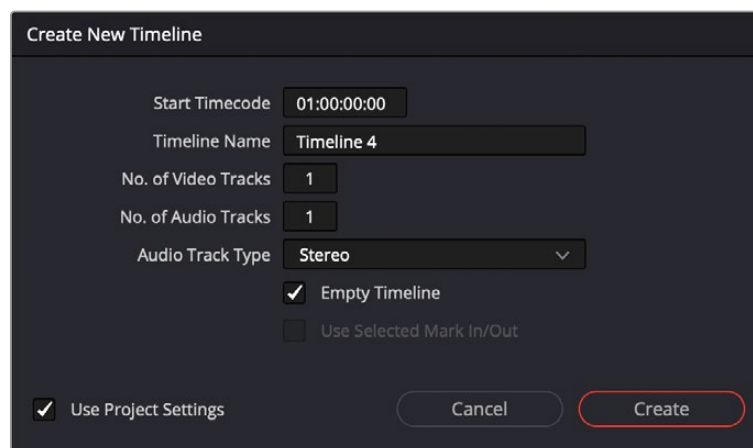
Creating and Duplicating Timelines

If you're not importing a project that's been edited elsewhere, you can create a new timeline to cut together a new edit from scratch, to use to assemble clips for use in a Fusion composition, to grade a set of dailies, or to put together an audio program within the Fairlight page. When you create a new timeline, you can either create a timeline that contains all clips found in the Media Pool to quickly create a big batch of offline dailies, or an empty timeline that's ready for you to add specific clips to. Timelines you create are stored in the currently selected Media Pool bin.

If you'd like an easy way to browse all the timelines in your project at once, regardless of their diverse locations, you can enable the "Smart Bin for Timelines" option, which is in the Editing panel of the User Preferences. This creates a Smart Bin in the Bin list of the Media Pool that filters all timelines in your project, making it easy to see all your timelines without altering your original organization.

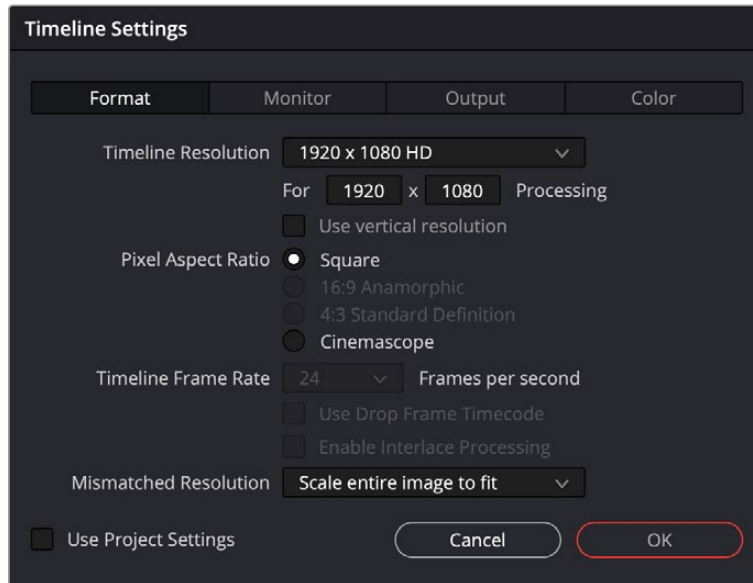
Individual Timeline Settings for Format, Monitoring, Output, and Color

When you create a new timeline, there are a number of ways you can customize it, but by default it will mirror the current project-wide timeline settings for resolution, frame rate, and other format and monitoring parameters.



The Create New Timeline dialog with standard customization settings

However, you have the option of creating separate timelines with individual Format, Monitoring, Output Sizing, and Color Management settings for situations where you need to set up multiple timelines to create multiple deliverables with different resolutions, pixel aspect ratios, frame rates, monitoring, or output scaling options, and color management than the overall project, including "Mismatched Resolution Files" settings. To choose individual settings, uncheck the Use Project Settings box in the New Timeline dialog, and additional controls will appear.



Clicking Use Custom Settings exposes additional panels for individualized timeline settings

Once you've created a timeline with individual settings, you can edit its settings by right-clicking that timeline in the Media Pool, and choose Timelines > Timeline Settings from the contextual menu. An Edit Timeline dialog appears, with separate panels for Format, Monitor, Output, and Color settings that you can choose. You can also click Use Project Settings to have that timeline use the project-wide timeline settings instead.

It is also possible to change the starting timecode for multiple timelines at the same time using the custom timeline settings.

To change the start timecode for multiple timelines at once:

- 1 Select the Timelines whose timecode you want to modify.
- 2 Right-click on one of the selected Timelines and choose Timelines > Starting Timecode from the contextual menu.
- 3 Enter the new timecode in the Time Start field.
- 4 Click on OK.

Creating Blank and Stringout Timelines

If you're cutting a new video or audio program, you'll usually want a blank timeline. However, the same command can be used to create stringout timelines when putting together dailies by turning the "Empty Timeline" checkbox off.

To create a new blank timeline:

- 1 (Optional) Select or create a folder in the Bin list in which to put the new timeline.
- 2 Do one of the following:
 - Choose File > New Timeline (Command-N).
 - Right-click within the Media Pool, and choose Timelines > Create New Timeline.
- 3 When the New Timeline Options window opens, set the following options:

Start Timecode: You can change the Start Timecode if a specific start time is required.

Timeline Name: Enter a name into the Timeline Name field.

No. of Video Tracks: Enter how many video tracks you want to have. You can also drag within this field to adjust the number of video tracks with a virtual slider.

Use Fairlight Preset: If this box is checked, it creates a timeline with pre-assigned audio tracks using a previously created Fairlight Configuration preset. A drop-down menu then appears, allowing you to select the specific preset for the Timeline. The preset is used in lieu of the No. Of Audio Tracks setting below. You can create Fairlight Configuration presets using the Fairlight Presets Library, available from the Fairlight menu.

For more information, see *Chapter 171, “Setting Up Tracks, Busses, and Patching.”* in the DaVinci Resolve Manual. If you have no Fairlight Configuration presets saved, this option will not appear.

No. of Audio Tracks: Enter how many audio tracks you want to have. You can also drag within this field to adjust the number of audio tracks with a virtual slider.

Audio Track Type: Choose the channel mapping you want the new audio tracks to use.

Empty Timeline: Checked by default, this sets new timelines to be created empty. If you turn off the Empty Timeline checkbox, the new Timeline that’s created will contain all media found in every bin within the Media Pool, effectively creating a stringout of everything you’ve imported.

Use Selected Mark In/Out: Only available when “Empty Timeline” is turned off. When you turn this checkbox on, each clip’s duration in the new Timeline is defined by the In and Out points saved within each clip. If there are no In/Out points in a clip, the clip’s entire duration is used.

Use Custom Settings: Click this button if you want to expose the Format, Monitor, and Output tabs that can expose unique settings for each timeline.

- 4 Click Create New Timeline.

A new timeline is created. If necessary, you can duplicate an existing timeline in order to alter an edit or create an alternate grade.

TIP: If you’re going to be creating several new timelines with a specific set of parameters, you can open the User pane of the Preferences and edit the New Timeline Settings, found in the Editing panel. This will define new presets that populate the New Timeline Options window from that point forward.

Creating Timelines by Drag and Drop

When you first create a new project, no timeline inhabits the Timeline Editor, and you have an opportunity to create a new timeline by drag and drop.

To create a timeline by dragging and dropping a clip:

Drag any clip into the empty Timeline Editor area underneath the Viewers on the Edit page, and a new timeline will automatically be created.

Creating Timelines From Bins and Selections

The “Create Timeline Using Bin” and “Create Timeline Using Selected Clips” commands let you quickly assemble a timeline using the contents of the Media Pool, using whatever In and Out points have been added to each clip, and using the sort order of the enclosing bin to determine the order in which the clips will be assembled.

TIP: These commands are especially useful for putting together quick assembly edits if you have metadata-rich media with scene, shot, and take information that you can use to sort clips into the proper order, and In and Out points that you’ve already logged.

To create a timeline using the full contents of a bin:

- 1 (Optional) Put the Media Pool into List mode, set In and Out points for each clip in your Bin, and sort the Media Pool by the column that will put all clips in the order you want them to be assembled.
- 2 Right-click the bin in the Bin list, and choose Create Timeline Using Bin.
- 3 Type the name of the new timeline in the New Timeline Properties dialog. If you want to use the In and Out points of each clip, make sure “Use Selected Mark In/Out” is checked, and click Create New Timeline.

To create a timeline using manually selected clips:

- 1 (Optional) Put the Media Pool into List mode, set In and Out points for each clip in your Bin, and sort the Media Pool by the column that will put all clips in the order you want them to be assembled.
- 2 Select one or more clips you want to assemble into a new timeline.
- 3 Right-click one of the selected clips, and choose Create Timeline Using Selected Clips.
- 4 Type the name of the new timeline in the New Timeline Properties dialog. If you want to use the In and Out points of each clip, make sure “Use Selected Mark In/Out” is checked, and click Create New Timeline. By default, Audio Track Type is set to “Based on selected media,” so the timeline audio tracks reflect the track mapping of the clips you’ve selected, but you can manually choose other specific mappings if you need to.

Creating Timelines Using an IMF or DCP Composition Playlist (CPL)

You can create a timeline in DaVinci Resolve that exactly replicates the Composition Playlist (CPL) of a DCP or IMF package. This is currently a DaVinci Resolve Studio only feature.

To create a timeline using a Composition Playlist (CPL):

- 1 Import an IMF or DCP package into the Media Pool, like any other piece of media.
- 2 Right-click on the imported clip and choose “Create New Timeline Using Composition Playlist” from the contextual menu.
- 3 In the New Timeline dialog box, choose a specific CPL from the package in the “Composition Playlist” drop-down menu.
- 4 Make any other normal new Timeline adjustments you may need (such as Resolution, Aspect Ratio, etc.), then click the “Create” button.

Duplicating Timelines

You can also duplicate existing timelines in preparation for saving a copy prior to making modifications, or as a starting point for a different version of your content.

To duplicate a Timeline, do one of the following:

- Select a timeline in the Media Pool, and choose Edit > Duplicate Timeline. The duplicate timeline appears with “copy” appended to the name.
- Right-click a timeline in the Media Pool, and choose Duplicate Timeline from the contextual menu.

Disabling Timelines

You can disable/enable timelines in the Media Pool for both performance and organizational purposes. This is particularly useful for editors who like to maintain a history of a program’s editing via an ongoing series of periodically duplicated timelines. Since having a large number of timelines within a single project file can affect performance, having the ability to disable timelines lets you maintain these backup/alternate timelines without any penalty.

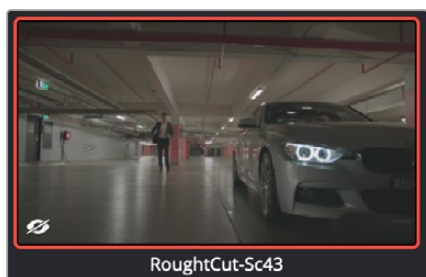
Disabled timelines are never loaded into RAM, have no effect on the speed at which a project opens, saves, exports, or loads, and have no effect on program performance. A disabled timeline also hides the timeline from the viewer drop-down menus throughout the program. Disabled timelines are still visible in the Media Pool, but have a crossed out eye icon in the lower left to show the status. A disabled timeline cannot be opened in any page of DaVinci Resolve.

To disable a timeline:

- Select the Timeline, right-click on it and choose “Disable Timeline” from the drop-down menu.

To enable a timeline:

- Select the Timeline, right-click on it and choose “Enable Timeline” from the drop-down menu.



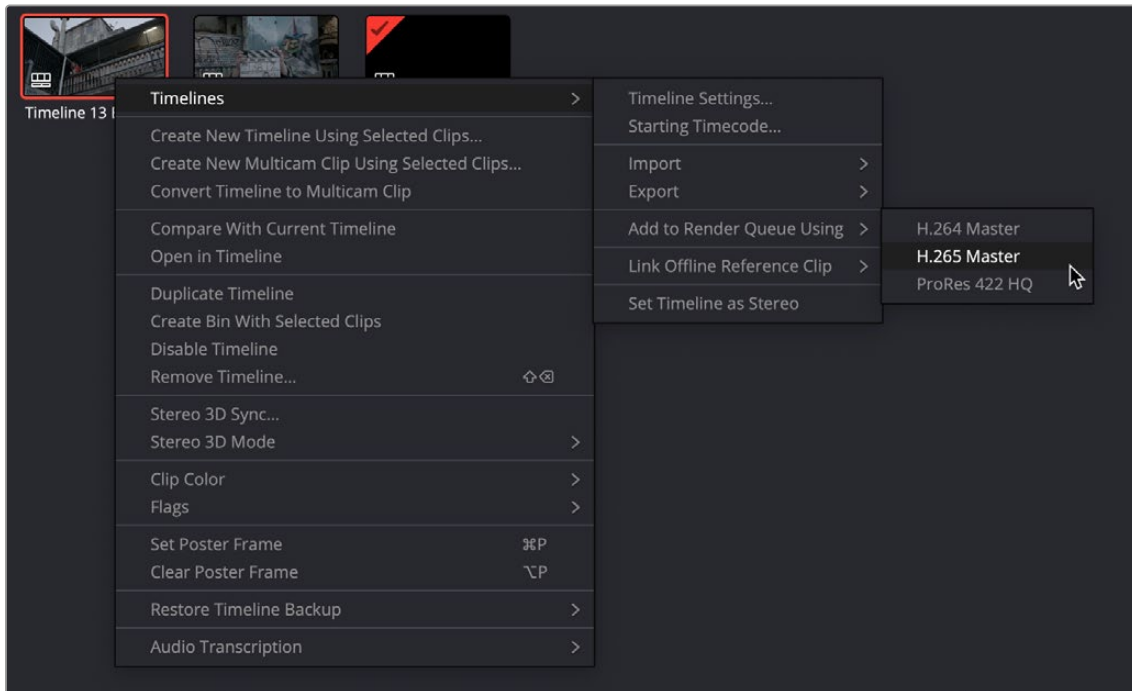
The crossed-out eye in the lower left of the thumbnail indicates this Timeline is disabled.

TIP: Enabled and Disabled timelines can be easily grouped and organized across the entire project by creating a Smart Bin with the MediaPool Properties > Timeline fields, and choosing “is Enabled/Disabled.”

Adding Media Pool Timelines Directly to the Render Queue

You can add a timeline directly to the Deliver page’s Render Queue by right-clicking on the timeline and hovering over the Add to Render Queue Using, and making a selection from the list of presets. You can add your own presets to this list by saving a new custom preset in the Deliver page.

You will then be prompted for the location to render the file. The timeline will be added to the Render Queue in the Deliver page but will not start rendering until the Render All button is pressed.



Select a Preset from the “Add to Render Queue Using” menu option to directly send the timeline to the Render Queue

Import and Export DaVinci Resolve Timelines (.drt)

You can export and import individual timelines from one DaVinci Resolve project into another previously existing DaVinci Resolve project, allowing you to pass timelines quickly between projects and workstations, without creating additional imported project files. Just the timeline and its associated clip information are exported; none of the actual media files are included.

To export a timeline from the Media Pool:

- 1 Select a timeline from the Media Pool.
- 2 Choose File > Export > Export AAF, XML, DRT (Shift-Command-O).
- 3 Choose “DaVinci Resolve Timeline Files (*.drt)” from the format options popup in the file system dialog.
- 4 Choose where to save the DaVinci Resolve Timeline file (.drt) in the file system dialog, and click Save.

To export multiple timelines from the Media Pool:

- 1 Select multiple timelines from the Media Pool.
- 2 Right-click on a timeline and choose Timelines > Export > DaVinci Resolve Timeline Files.
- 3 Select the folder where you wish to save them from your file browser.

Unlike exporting individual timelines, when exporting multiple timelines, you will not be able to rename the .drt files on export. They will retain the timeline name they have in the Media Pool.

To import a timeline into the Media Pool:

- 1 Choose a bin in the Media Pool in which you want the imported timeline to be saved.
- 2 Do one of the following:
 - Choose File > Import Timeline > Import AAF, XML, DRT (Shift-Command-I), then Select a DaVinci Resolve Timeline file (.drt) from the file system dialog, and click Open.
 - Double click the .drt file in your file system.

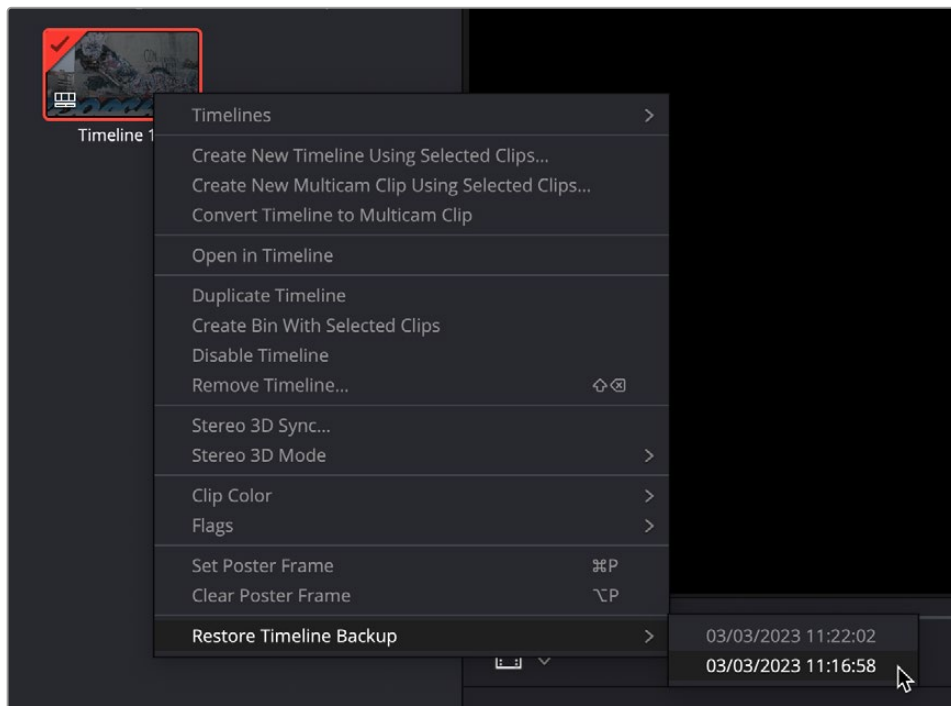
The timeline will appear in the Media Pool, along with all of the clips associated with it, including any media sync information. Any timelines imported this way will have the word “import” appended to their name, to avoid duplicate names. The imported timeline will be automatically conformed to corresponding media that’s already in the Media Pool. However, if the timeline has been moved to another computer, you may have to reimport or relink missing or offline media in to bring the imported timeline fully online.

NOTE: Only a single timeline can be imported and exported at a time using this method. To import or export multiple timelines, use the Import/Export Bin function instead.

Backing Up and Restoring Timelines

Turning on the Timeline Backups checkbox in the Project Save and Load panel of the User Preferences enables DaVinci Resolve to save multiple backups of a timeline at periodic intervals, using a method that’s analogous to a GFS (grandfather-father-son) backup scheme. This can be done regardless of whether or not Live Save is turned on.

If you want to revert to a previous backup of a timeline, simply right-click on the Timeline in the Media Pool, select Restore Timeline Backup from the contextual menu, and choose the backup from the list of options. Backups are organized by date and time, making it easy to find the specific timeline you want to restore.



Restoring a timeline backup in the Media Pool

Restoring a timeline backup does not overwrite your current timeline. Instead the selected backup will be brought into the Media Pool as a new timeline with the name “Backup” appended to it.

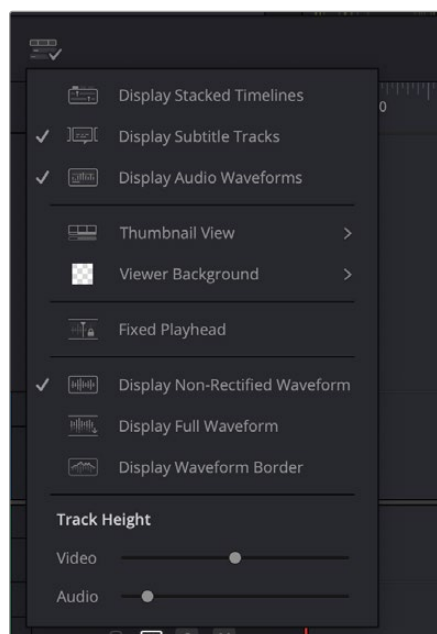
Timeline backups are only saved when changes have been made to a project. If DaVinci Resolve sits idle for any period of time, such as when your smart watch tells you to go outside and walk around the block, no additional project backups are saved, preventing DaVinci Resolve from overwriting useful backups with unnecessary ones.

Three fields let you specify how often to save a new backup, while the fourth lets you choose where the backups will be saved. These settings apply to both Project and Timeline backups.

- **Perform backups every X minutes:** The first field specifies how often to save a new backup within the last hour you’ve worked. By default, a new backup is saved every 10 minutes, resulting in six backups within the last hour. Once an hour of working has passed, an hourly backup is saved and the per-minute backups begin to be discarded on a first in, first out basis. By default, this means that you’ll only ever have six backups at a time that represent the last hour’s worth of work.
- **Hourly backups for the past X hours:** The second field specifies how many hourly project backups you want to save. By default, two hourly backups will be saved for the current day. Past that number, hourly backups will begin to be discarded on a first in, first out basis.
- **Daily backups for the past X days:** The third field specifies for how many days you want to save backups. The very last backup saved on any given day is preserved as the daily backup for that day, and by default daily backups are only saved for two days. Past that number, daily backups will begin to be discarded on a first in, first out basis. If you’re working on a project over a longer stretch of time, you can always raise this number.
- **Project backup location:** Click the Browse button to choose a location for these backups to be saved. By default they’re saved to a “ProjectBackup” directory on your scratch disk, although you could change this to a volume that better fits into your data backup methodology. This folder contains both Project and Timeline backups.

Timeline View Options

As you’re working on an edit, it can often be useful to modify the appearance of the Timeline, changing the height of video or audio clips, choosing whether audio waveforms are drawn or not, and so on. Using the Timeline View Options drop-down at the far left of the Timeline, you can make these kinds of changes as you work.



The Timeline View Options drop-down

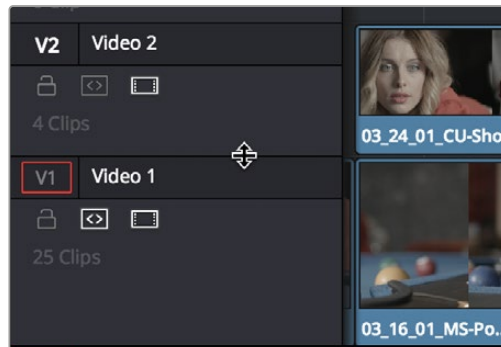
You have the following options:

- **Timeline View Options:** Three buttons let you choose to show or hide specific Timeline interface elements, including the following:
 - Display Stacked Timelines:** This option lets you show stacked timelines for simultaneous display of timelines one on top of another.
 - Display Subtitle Tracks:** Lets you display or hide the subtitle tracks region of the Timeline. Hiding subtitle tracks does not disable subtitle display; to do that you must disable the currently displayed subtitle track.
 - Display Audio Waveforms:** Lets you turn audio waveform viewing off and on. When Audio Waveform is turned off, audio tracks are minimized.
- **Thumbnail View Options:** Three buttons let you choose the overall appearance of the video track. From left to right:
 - Filmstrip:** Displays each clip as a series of frames along its length. The number of frames displayed depends on the current zoom level of the clip in the Timeline.
 - Thumbnail:** Displays each clip as a solid color with a thumbnail image of the clip's In point at the beginning of the clip and a thumbnail image of the Out point at the end. The thumbnails displayed depend on the current zoom level and track height of the clip in the Timeline. If the clip does not have enough room for both, only the In point thumbnail will appear.
 - None:** Displays each clip as a solid color along its length.
- **Viewer Background:** Four options: Checkerboard, Black, Gray, or Alert Red let you determine what color the transparent background will be in the Viewer.
- **Fixed Playhead:** This option lets you toggle between a fixed playhead, keeping the playhead static in the middle of the timeline, while the tracks on the timeline move underneath it, or Free Playhead, where the playhead travels along the timeline as it is played back.

When using the Fixed Playhead option, you can choose where to set the location of the Fixed Playhead in the timeline, rather than just have it in the center. To do so, hover the pointer over the top of the playhead in the Timeline Ruler and drag it to the new location you want to use.

Moving the Fixed Playhead can be useful in increasing the amount of timeline working area for clips before or after the current position.
- **Audio View Options:** Three buttons govern the look of audio waveforms in the Timeline, when visible.
 - Display Non-Rectified Waveform:** Lets you toggle between the waveform being drawn from the bottom of the audio track up, or centered and mirrored about itself.
 - Display Full Waveform:** Hides the divider bar that keeps the waveform separate from the file name area of each audio clip, so the waveform occupies the full space of each audio bar in the Timeline.
 - Display Waveform Border:** Draws a dark border around the edges of each waveform to make them easier to see.
- **Video track height slider:** Lets you resize the size of all video tracks at once, independently of the audio tracks.
- **Audio track height slider:** Lets you resize the size of all audio tracks at once, independently of the video tracks.

In addition, any track in the Timeline can be individually resized by dragging its top divider in the Track Header area. Track heights in the Edit page are independent of the Thumbnail and Waveform view settings in the Timeline View options. Previously, specific Timeline viewing options such as Filmstrip or Thumbnail view had minimum track height settings. Now you can freely change track height no matter what options you've chosen, and resizing one or more tracks below the minimum height for filmstrips or thumbnails automatically collapses those tracks into Simple view to avoid clutter.



Resizing an individual Timeline track by dragging its top border in the Track Header

Modifying Timeline Tracks

When you're getting ready to edit clips into the Timeline, you need to make sure you've got enough tracks to do the job. The following procedures cover the different methods available for adding, removing, and rearranging tracks as you work. These commands are all available via the contextual menu that appears when you right-click anywhere in the Timeline header area (the header of the Timeline is the area to the left where each track's various buttons and controls are located).

Keyboard Shortcuts can be assigned for track header context menu actions to add, delete, move, change color, and link tracks. Tracks can be selected in the Track index, or the Edit and Fairlight timelines. For example, you can assign a keyboard shortcut to add a new Audio Track in the Edit page by navigating to Commands > Edit Timeline > Track Controls > Add Track > Audio in the Keyboard Customization window.

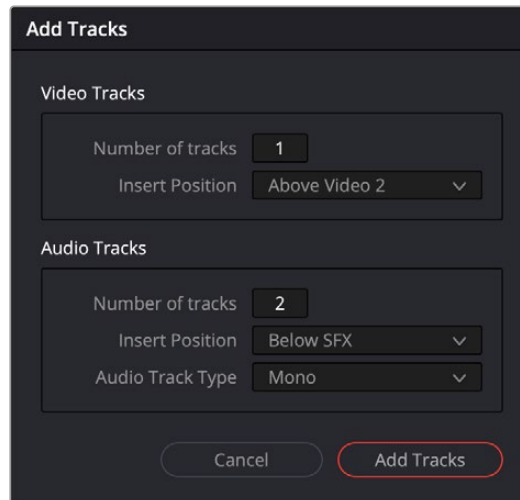
Methods for adding, deleting, and rearranging tracks:

- **To add a track to the Timeline:** Right-click on a Track Timeline header and choose Add Track. A new video track will be created immediately above the track you right-clicked on. A new audio track will be created immediately below the track you right-clicked on. If you add an audio track, you can choose what type of channel mapping you want.

For more information about audio track channel mappings, see Chapter 45, "Working with Audio in the Edit Page."

- **To add multiple tracks to the Timeline at a specific position:** Right-click anywhere in the Timeline header and choose Add Tracks. When the Add Tracks dialog appears, choose the number of video and audio tracks you want to add, choose the position you want to insert the tracks above or below, and choose the Audio Track Type you want to add if you're adding audio tracks. When you're done, click "Add Tracks."
- **To delete a track from the Timeline:** Right-click within a track's Timeline header and choose Delete Track. If there are clips on a track you remove, they are deleted from the Timeline.

- **To delete all unused tracks in the Timeline:** Right-click anywhere in the track header area and choose Delete Empty Tracks. All tracks without clips will be deleted at once.
- **To move tracks and the clips on them up and down:** Right-click within a track's Timeline header and choose Move Track Up or Move Track Down from the contextual menu. That track, along with all clips on it, will be moved up or down relative to the other tracks in the Timeline. You can also rearrange track order from the Tracks Index. Simply click on the track in the # field and drag the track to its new position.



The Add Tracks Dialog lets you add multiple tracks at once and set their locations in the timeline.

Naming Timeline Tracks

If you're a stickler for organization, you can also name the tracks on a timeline to identify its purpose.

To rename a track:

- 1 Make sure the Timeline View Options are set to either the Filmstrip view or the Thumbnail view, and that audio tracks are tall enough, so that track names are visible.
- 2 To edit the name of any track, click the default "Video X" or "Audio X" track name to select it, then type your preferred name and press the Return key.

Using the Tracks Index

The Tracks Index presents an easy to use track list that lets you view and control all your track headers in one place. It also gives more audio information than the track header section of the Timeline. The Tracks Index is especially useful for controlling projects with a large number of video and audio tracks. It is accessed by clicking on the Index pane and selecting the Tracks tab.

The resulting list shows all the current tracks in your timeline and their position and attributes. These are all modifiable right inside the Track Index.

#	Name	Track Controls	Format	Monitor	ADC
V3	Adjustment Clips				
V2	Titles				
V1	Video 1				
A1	Dialog Mix		1.0	☐	☒
A2	SFX		1.0	☒	☒
A3	Music		1.0	☐	☐
A4	Foley		2.0	☒	☒

The Tracks Index

Tracks Index Columns

Each track reveals several columns of information. These columns can be rearranged by dragging the column headers to the left or right, depending on what information is most important to you. Clicking on a track row selects the track for modification. Right-clicking on a track or tracks allows you to delete them from the contextual menu.

Using the Tracks Index for video tracks:

- **Color:** Shows a bar of the track's assigned color. You can right-click on the track to change its color.
- **#:** Shows the track's number.
- **Name:** Shows the track's name. This field is editable.
- **Track Controls:** Exposes the Lock, Auto Track Selector, and the Enable/Disable Video track toggle. All of these controls are active.

Using the Tracks Index for audio tracks:

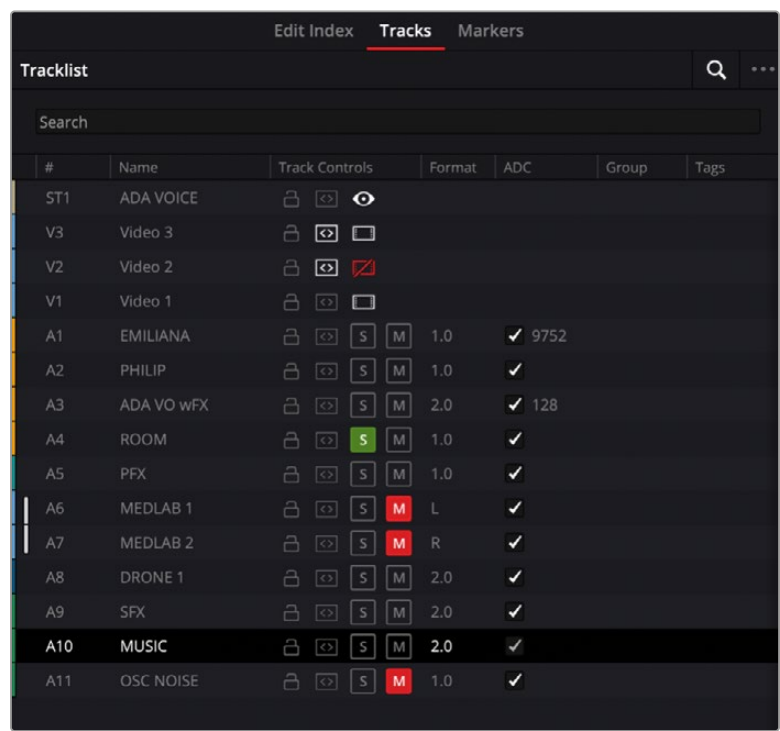
- **Color:** Shows a bar of the track's assigned color. You can right-click on the track to change its color.
- **#:** Shows the track's number.
- **Name:** Shows the track's name. This field is editable.
- **Track Controls:** Exposes the Lock, Auto Track Selector, Solo, and Mute controls. All of these controls are active.
- **Format:** Shows the audio track type. Right-clicking on this field lets you change the audio track type.
- **ADC:** Shows the current amount of Automatic Delay Compensation on the track. Clicking on the checkbox toggles the automatic delay compensation on or off.
- **VCA:** Shows the track's VCA group number.
- **Tags:** Shows any tags assigned to the track.

Using the Tracks Index for subtitle tracks:

- **Color:** Shows a bar of the track's assigned color. You can right-click on the track to change its color.
- **#:** Shows the track's number.
- **Name:** Shows the track's name. This field is editable.
- **Track Controls:** Exposes the Lock, Auto Track Selector, and the Enable/Disable Subtitle toggle. All of these controls are active.

Using the Tracks Index options:

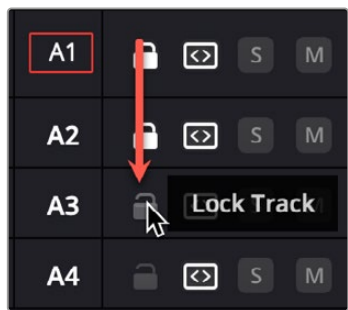
- **To search for a specific track:** Click on the search icon (magnifying glass), and type in your search terms.
- **To select which type of tracks are included in the Tracks Index:** Click on the Option menu to choose which type of tracks will be included in the list. The options are: Video, Audio, and Subtitle.



The Tracker Index lets you control all your track headers in one place.

Toggle Track Header Control States by Clicking and Dragging

You can change multiple Track Header controls in the Edit page (Lock Track, Visibility, Auto Track Selection, etc.) by simply clicking on an icon and dragging your pointer up or down across the track headers. You can do this for one type of control at a time and not across Audio and Video tracks.



Clicking on the Lock icon on A1, then dragging the pointer down to A3, will lock all audio tracks in between.

Using Timeline Snapping and Zooming

When preparing to make an edit into the Timeline, you can set the snapping and zoom controls to whichever state is most useful for the operation you need to perform. For example, if you're editing an insert shot into a rapid-fire section of edits in the Timeline, you may want to zoom in to better see the exact place where you want to place the incoming clip. Since there are many edits at that point, disabling snapping might make it useful to avoid having the clip jump to the nearest edit point if you need to move the incoming clip to a very specific frame.

Playhead Snapping:



To turn clip and playhead snapping on and off: Click the Snapping button in the toolbar, or press N. When snapping is turned on, the In and Out points and markers of clips all snap to one another and to the playhead. You can also press N to temporarily turn snapping on or off while dragging a clip in the Timeline, or while scrubbing the playhead using the pointer (snapping reverts to its previous state when you finish the operation).

Timeline Zoom presets:



From left to right: Full Extent, Detail, and Custom Zoom controls

	Full Extent Zoom: This mode dynamically adjusts the zoom level to encompass the whole Timeline as you add or remove clips. As your timeline grows by adding clips, the zoom level decreases automatically to fit the new clips into your Timeline Viewer. Conversely, the zoom level will increase automatically as you remove clips.
	Detail Zoom: This setting zooms the Timeline in on the Playhead to the frame level to quickly make fine timing adjustments.
	Custom Zoom: Zooms the Timeline to the level selected by the Zoom Slider to its immediate right.
	Zoom Slider: Drag the zoom slider to the left to zoom out, and right to zoom in. You can also press Command-Minus (–) and Command-Equal (=) to zoom out and in. Either way, zooming is always centered on the current position of the playhead, even if the playhead is off screen.
	To frame every clip into the width of the Timeline: Press Shift-Z. This is a toggle, so pressing Shift-Z frames your whole edited sequence to the width of the Timeline, and then pressing Shift-Z again returns the Timeline to whatever level of zoom you were using previously. Using this keyboard shortcut makes it really easy to navigate the Timeline when you're zoomed in, as you can press Shift-Z, move the playhead to another part of the Timeline you want to work on, and then press Shift-Z again to zoom back into the new location of the Timeline.

Zoom Around Mouse Pointer

You can set up the Edit Timeline zoom controls to stay centered on the pointer as you zoom in or out, instead of staying centered on the Timeline playhead as usual, by selecting View > Zoom Around Mouse Pointer. This can be helpful when navigating longer timelines.

With this option enabled, zooming in and out of the Timeline using the scroll control of your pointing device while holding the Option key down keeps the Timeline centered on the pointer. Deselect this option to return to the behavior of only zooming in or out centered on the playhead position.

Scrolling Through the Timeline

When you're zoomed into the Timeline, there are several methods you can use to scroll around as you work.

- You can scroll left and right, or up and down, by dragging the Timeline's horizontal and vertical scroll bars.
- You can scroll left and right through the Timeline, one "page" at a time at the current zoom level, by selecting Playback > Next/Previous > Timeline Page. If you use these often, you can assign them a keyboard shortcut.
- You can also scroll up and down the tracks of the Timeline using the scroll wheel, scroll ball, or scroll gestures of your mouse, trackball, trackpad, or other pointing device.
- You can also scroll within the Timeline by middle-clicking and dragging in any direction, which works the same as panning around a viewer that you've zoomed into. This frees you from having to use the scroll bars as you move around your edited clips.

If you scroll past the position of the playhead, a small playhead indicator appears in the bottom scroll bar to let you know where it is relative to the entire duration of your edited sequence.



A small indicator shows the position of the playhead if it's outside of the visible area of the Timeline.

Scroll Wheel Controls on Timeline

Additional scroll wheel and key presses can be used to manipulate the Timeline on the Edit page.

To manipulate the Timeline, rotate the scroll wheel while holding down:

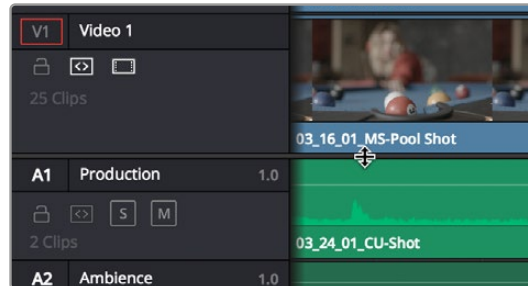
Shift: Expands or contracts all the Video or Audio track heights, depending on what section the pointer is located at when you scroll.

Option: Zooms in and out the view of the Timeline.

Command: Navigates forward and backward in the Timeline.

Resizing the Timeline's Video and Audio Track Regions

If you need to see more of the video or audio tracks in the available area of the Timeline, you can drag the horizontal divider that separates the audio and video tracks up or down to create room where you need it.



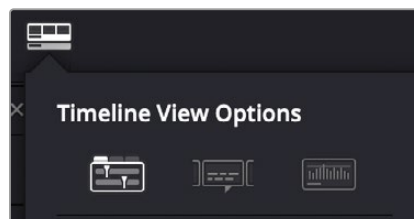
Dragging the Timeline center divider to make more room for audio or video tracks

Tabbed and Stacked Timelines

The Timeline now supports the option to have tabs that let you browse multiple timelines quickly. With tabbed timeline browsing enabled, a second option lets you open up stacked timelines to simultaneously display two (or more) timelines one on top of another.

Tabbed Timelines

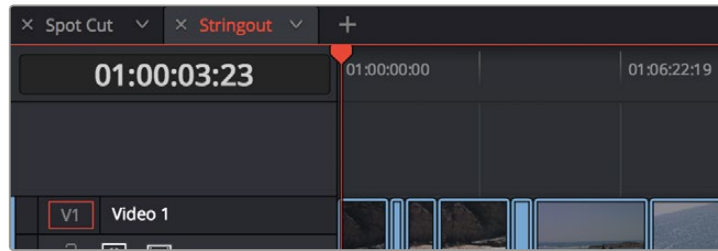
The Timeline View Options menu in the toolbar has a button that lets you enable tabbed browsing and the stacking of timelines.



A button in the Timeline View Options enables tabbed timelines.

When you first turn this on, a Timeline tab bar appears above the Timeline, displaying a tab for the currently open timeline that contains a Close button and a Timeline drop-down menu. Once you enable Tab mode, opening another timeline from the Media Pool opens it into a new tab.

To the right of the currently existing tabs, an Add Tab button lets you create additional tabs that default to "Select Timeline." Click any tab's drop-down menu to choose which timeline to display in that tab.



Tabs above the Timeline Editor let you switch among multiple timelines quickly

Right-clicking a tab opens a contextual menu that gives access to several commonly used Timeline functions.

Tabbed timeline contextual menu options:

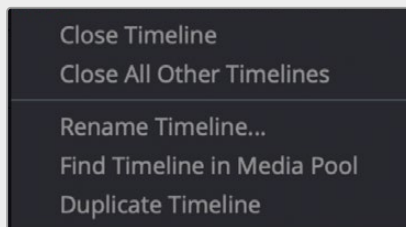
Close Timeline: Closes the current tab's timeline, and removes the tab. The same as clicking on the "x" inside the tab.

Close All Other Timelines: Closes all open timelines and tabs except for the one that you right clicked and chose this option on.

Rename Timeline: Opens the tab's text editing field, allowing you to change the name of the timeline inside the tab. This will change the timeline's name across the project in the Media Pool as well.

Find Timeline in Media Pool: Opens the bin where the timeline is stored in the Media Pool, and highlights the timeline in orange.

Duplicate Timeline: Creates a duplicate of the selected timeline in the same bin, with the word "copy" appended to the timeline name. It also automatically opens the copied timeline in a new tab.



Right-clicking on a tab will show the timeline options.

Methods of working with tabbed timelines:

- Click any tab to switch to that timeline.
- Use the drop-down menu within any tab to switch that tab to display another timeline from the Media Pool. Each tab's drop-down menu shows all timelines within that project, in alphabetical order, but a timeline can only be open in one tab or stack at a time.
- Drag any tab left or right to rearrange the order of timeline tabs.
- Click any tab's Close button to close that timeline and remove that tab.
- Middle-click any tab to close that timeline and remove that tab.

Stacked Timelines

While tabbed browsing is turned on, an Add Timeline button appears on the far right of the tab bar that enables you to stack two (or more) timelines one on top of another. This lets you have two (or more) timelines open at the same time, making it easy to edit clips from one timeline to another.

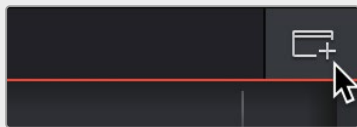
A good example of when this is useful is when you've created a timeline that contains a stringout of selects from a particular interview. You can stack two Timeline Editors, one on top of another, and then open the Selects Timeline at the top and the Timeline you're editing into at the bottom. With this arrangement, it's easy to play through the top timeline to find clips you want to use, to drag and drop into the bottom timeline to edit into your program.



The button for closing a stacked timeline

To enable or disable stacked timelines:

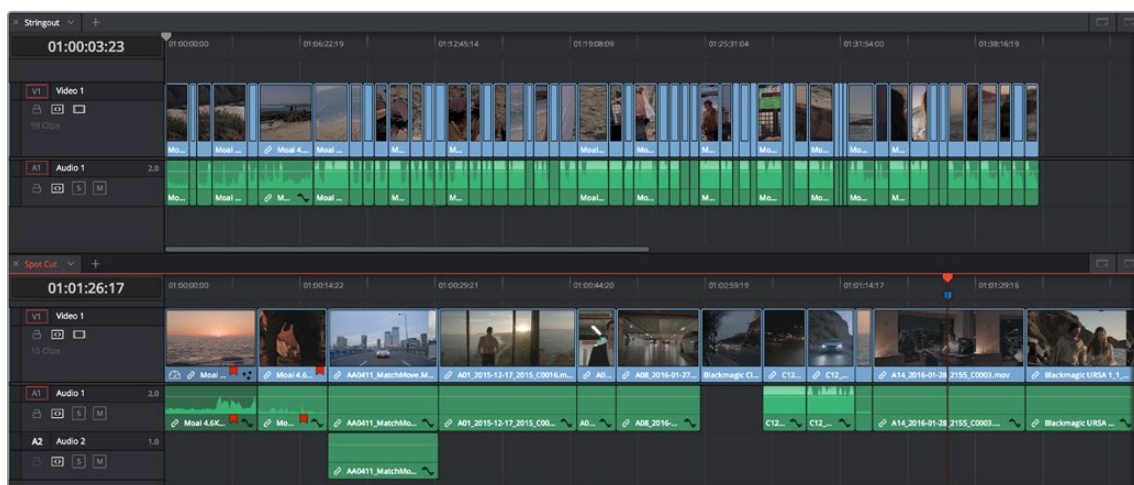
Click the Add Timeline button at the right of the Timeline tab bar.



The button for adding a stacked timeline

Once you've enabled stacked timelines, each timeline has its own tab bar and an orange underline shows which timeline is currently selected.

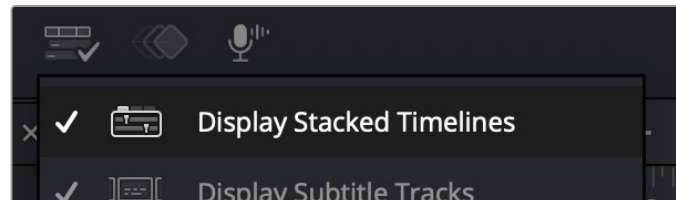
At the right of each Timeline tab bar, a Close Timeline button appears next to the Add Timeline button, which lets you close any timeline and remove that timeline browsing area from the stack.



Two timelines stacked on top of one another

View and Edit Dual Timelines Using the Source Viewer

You can now work on two timelines simultaneously in the Edit page, one loaded in the Source Viewer and one in the Timeline Viewer. This allows you to easily compare two timelines or copy content between one timeline to another.



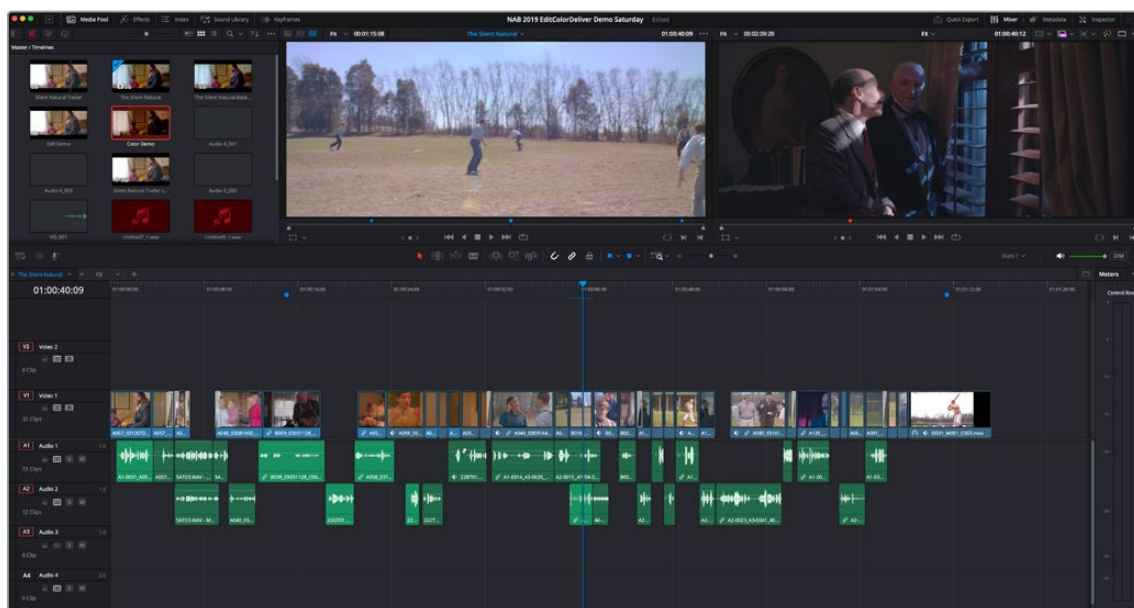
Enable Display Stacked Timelines in the Timeline View options to use dual timelines

To get the best use from this feature, you should first enable Display Stacked Timelines in the Timeline View Options icon in the upper left of the timeline. Enabling this view will allow each timeline to appear in its own tab, and the Viewer will automatically switch focus, depending on which tab is selected.

Then, in order to load a timeline into the Source Viewer, right-click on a timeline in the Media Pool and choose “Open in Timeline with Source Viewer.” The timeline will appear in the Source Viewer, as a new tab with the name and playhead turning blue to help you keep track of what timeline is in what Viewer. The timeline in the Source Viewer is read only. You can switch between the timelines by clicking on the tabs, pressing Q, or by clicking in the appropriate Viewer.

You can change the timeline that appears in the Source Viewer either by right-clicking another timeline in the Media Pool and selecting the “Open in Timeline with Source Viewer” option, or by selecting one of the timelines in the dropdown menu at the top of the Source Viewer.

You can perform standard cut and copy operations from the read-only source timeline to the actual timeline, as well as drag from the Source Timeline viewer (left) into the Timeline viewer (right) to make an edit.



A dual timeline editing setup, with the Silent Natural timeline in the Source Viewer (blue), and the FX timeline in the Timeline Viewer. Switching tabs automatically focuses the appropriate Viewer.

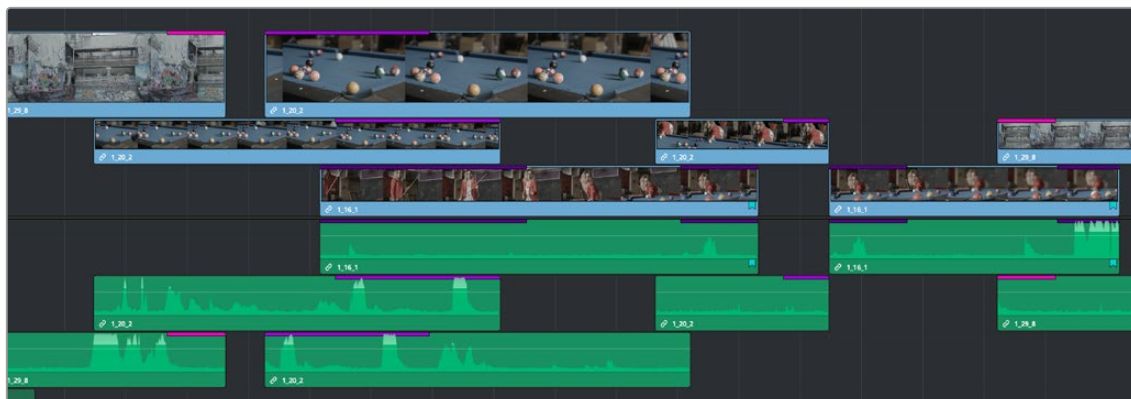
There are also keyboard shortcuts that make focusing between viewers that much quicker:

- To toggle between the Source and Timeline Viewers, press Q.
- To turn on and off the Source Timeline view, press Option-Q.

TIP: You can also right-click on a normal media clip and select Open in Timeline with Source Viewer to load that clip into the Viewer as a timeline. These “timelines” will be read only, but will otherwise allow the same timeline to timeline editing operations currently possible. This mode is useful for very long clips like interviews.

Duplicate Frame Detection

You can turn on Duplicate Frame Detection (often referred to as Dupe Detection) for clips in the Timeline by choosing View > Show Duplicate Frames. Doing so shows colored bars at the top of clips in the Timeline whenever a range of frames has been used more than once.



Duplicate Frame Detection shows colored bars where a range of frames is used more than once in a timeline.

Comparing Timelines

For instances where you're importing multiple versions of a timeline that's been edited in another application, or where you're working with multiple editors on different versions of the same Timeline in either collaborative mode or on multiple separate DaVinci Resolve installations, DaVinci Resolve provides a method of comparing two timelines with one another. Using the Timeline Comparison window, you can both see a visual comparison of which sections of two timelines differ, and you can derive a more traditional change list by opening up the Difference Index.

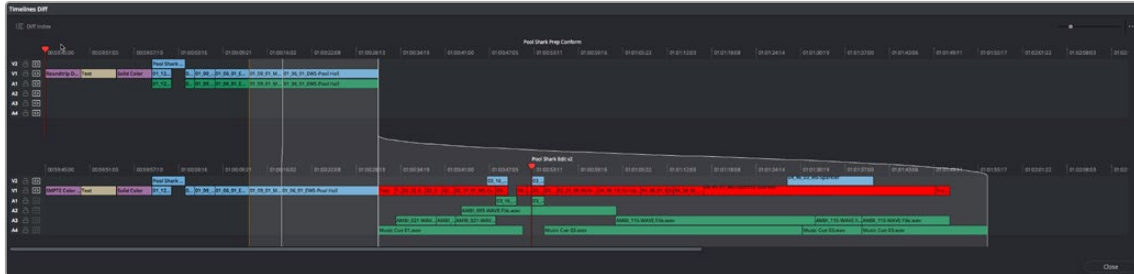
To compare two timelines:

- 1 Open the first timeline you want to compare.
- 2 Right-click a second timeline in the Media Pool, and choose Compare With Current Timeline.

A Timeline Comparison window appears, showing you the currently opened Timeline at the bottom and the Timeline you right-clicked at the top.

The Timeline Comparison Window

When you first open the Timeline Comparison window, the first thing you see is a pair of miniature timelines. The currently open Timeline appears at the bottom and the Timeline you right-clicked appears at the top.



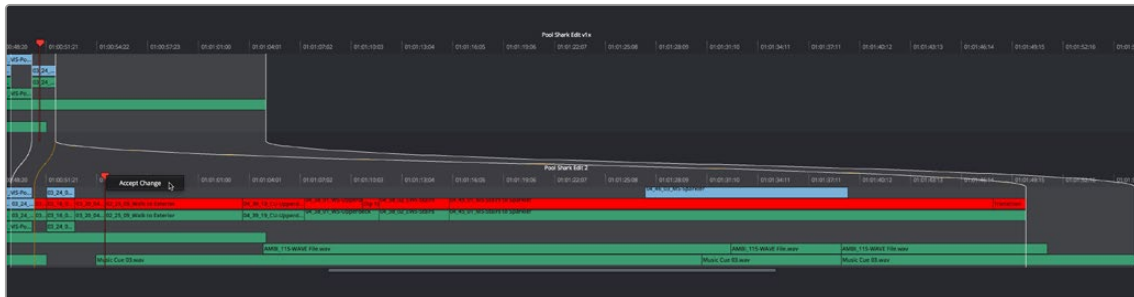
The Timeline Comparison window

Comparison Window Playhead Output

By default, the two playheads are ganged together, with the top playhead being displayed in the Source Viewer, and the bottom playhead being displayed in the Timeline Viewer. These playheads can be unganged if you want to compare different areas of both timelines, simply by turning off Sync Playheads in the option menu.

Highlighting Differences

Special highlights indicate sections of both timelines that are different. Individual changes are not individually highlighted, although they can be seen, on the premise that you're more interested in a section by section analysis of what your collaborating editor has been doing, for purposes of deciding whether to incorporate changes or reversions based on this comparison.



Each section of differences between the two timelines are highlighted.

When using this tool, you can change the bottom Timeline to match the top Timeline, on a section by section basis, by right-clicking a highlighted section and choosing Accept Change from the contextual menu. When you do this, the currently open Timeline is immediately changed to incorporate the altered section from the Timeline you're comparing to. If necessary, you can undo this.

The Change List

Clicking the Diff Index button opens the change list, which shows you a more conventional item by item comparison of the differences between the two timelines.

Timelines Diff

☰ Diff Index

#	In	Out	In	Out
1	01:00:49:20	01:00:51:02	01:00:49:20	01:00:49:20
2	01:00:51:02	01:00:52:12	01:00:49:20	01:00:51:06
3	01:00:52:12	01:00:52:12	01:00:51:06	01:01:50:12
4	01:00:52:12	01:01:05:02	01:01:50:12	01:01:57:01

V2

V1

A1

A2

A3

A4

A5

The change list of the Timeline Comparison window